

ST JOHNSBURY, VT, USA

WMO: 726140

Lat: 44.420N

Lon: 72.019W

Elev: 697

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 93-19

WBAN: 54742

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-7.3	-19.4	2.0	-9.8	-14.4	2.6	-5.9	15.1	38.2	12.7	25.3	1.4	350	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	86.4	70.2	83.4	68.6	80.9	66.9	73.1	81.7	71.3	79.4	69.7	77.0	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	114.9	76.5	68.8	108.6	74.8	67.0	102.0	73.3	37.1	81.6	35.5	79.6	34.2	77.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.3	8.4	-21.4	91.8	6.7	1.8	-26.2	93.1	-30.0	94.1	-33.8	95.1	-38.6	96.4
			-21.7	76.3	6.5	1.8	-26.4	77.6	-30.2	78.6	-33.9	79.6	-38.6	80.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	17.3	18.8	29.0	42.8	54.9	63.6	68.7	66.8	59.1	47.3	35.5	24.7
	DBStd	20.12	13.36	11.47	10.80	8.35	7.66	6.51	5.06	5.43	7.23	8.02	8.87	11.43
	HDD50	4209	1015	873	654	248	33	0	0	0	8	151	441	786
	HDD65	7937	1480	1293	1116	667	325	103	21	45	204	549	885	1249
	CDD50	2086	0	0	3	32	186	408	580	521	281	68	6	1
	CDD65	339	0	0	0	0	13	60	137	101	27	1	0	0
	CDH74	2992	0	0	3	29	212	616	1133	766	224	9	0	0
	CDH80	736	0	0	0	4	49	173	293	169	48	0	0	0
(14) Wind	WSAvg	2.2	2.5	2.4	2.8	3.4	2.8	2.0	1.4	1.3	1.5	1.7	2.3	1.9
(15) Precipitation	PrecAvg	37.6	2.5	2.0	2.5	2.8	3.2	3.7	3.8	3.9	3.2	3.3	3.3	3.1
	PrecMax	49.4	6.8	4.9	4.4	6.2	7.9	9.6	7.2	7.7	8.6	8.6	7.0	7.9
	PrecMin	26.8	0.4	0.4	0.6	0.6	0.6	1.2	1.2	0.9	0.8	0.4	1.5	1.1
	PrecStd	5.7	1.4	0.9	0.9	1.2	1.6	1.7	1.4	1.8	1.5	1.8	1.2	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	51.0	65.7	77.6	86.0	90.2	90.1	88.5	85.8	75.2	63.9	59.5
		MCWB	49.2	42.9	52.0	57.5	66.0	71.6	73.3	72.4	71.7	62.6	56.7	39.5
	2%	DB	45.4	43.7	54.7	70.9	79.9	84.6	86.1	84.1	79.8	69.5	58.1	48.1
		MCWB	38.1	38.1	44.5	54.7	62.0	68.5	70.6	69.8	67.4	58.6	51.6	41.8
	5%	DB	39.1	38.5	48.3	64.5	75.0	81.0	83.2	81.4	75.8	64.5	53.3	43.0
		MCWB	35.8	34.9	40.4	51.3	59.1	65.7	68.8	68.2	65.3	56.3	47.6	38.8
	10%	DB	35.2	35.0	43.9	58.9	70.4	77.2	80.6	78.6	72.1	60.5	49.0	38.6
		MCWB	32.9	32.0	38.1	47.6	57.3	64.1	67.6	66.7	63.6	53.8	44.1	35.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	44.4	52.9	62.6	68.4	74.0	76.1	74.4	72.4	66.3	58.6	46.4
		MCDB	53.9	47.8	61.1	72.0	80.2	84.0	85.0	82.8	81.6	71.3	62.1	48.2
	2%	WB	40.3	38.6	46.0	57.1	65.8	71.5	73.3	72.2	69.8	61.9	53.3	42.4
		MCDB	42.6	42.2	52.4	67.9	75.4	80.8	81.8	80.1	76.8	66.3	57.3	47.7
	5%	WB	35.7	35.2	41.7	52.8	62.7	69.0	71.5	70.5	67.7	58.2	48.7	38.6
		MCDB	38.3	38.2	47.6	61.8	70.0	76.7	79.4	77.7	73.6	63.0	52.6	42.6
	10%	WB	32.9	32.5	38.3	49.3	59.7	66.5	69.8	68.9	65.2	54.9	44.4	35.5
		MCDB	35.5	34.8	43.1	58.3	67.5	73.5	77.3	75.4	70.5	59.4	47.9	38.4
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	20.3	19.9	20.8	22.3	21.1	20.4	20.7	20.6	18.1	15.3	14.7
		MCDBR	18.4	22.8	26.7	32.8	30.4	27.0	24.1	23.5	25.3	24.7	22.0	17.1
		MCWBR	15.6	17.6	17.4	18.4	14.6	12.4	10.6	10.8	13.2	15.3	16.5	13.3
	5% WB	MCDBR	17.7	20.9	23.7	28.4	23.5	22.6	20.1	20.0	21.2	20.6	19.2	15.6
		MCWBR	15.9	16.7	16.7	17.9	12.4	11.8	10.1	10.1	12.2	14.5	16.2	13.5
(40) Clear-Sky Solar Irradiance	taub		0.264	0.283	0.319	0.354	0.374	0.425	0.434	0.406	0.366	0.329	0.308	0.275
	taud		2.399	2.319	2.339	2.377	2.384	2.238	2.223	2.291	2.398	2.509	2.499	2.441
	Ebn at Noon		276	289	291	288	284	269	265	268	272	269	257	260
	Edh at Noon		23	30	34	36	37	43	43	39	32	25	22	21
(44) All-Sky Solar Radiation	RadAvg		454	728	1088	1353	1561	1703	1734	1570	1269	752	472	340
	RadStd		54	79	71	165	181	152	137	85	92	81	61	29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	N/A

Nomenclature: See separate page